CME 250:
Introduction to Machine Learning

Lecture 7:
Unsupervised Learning

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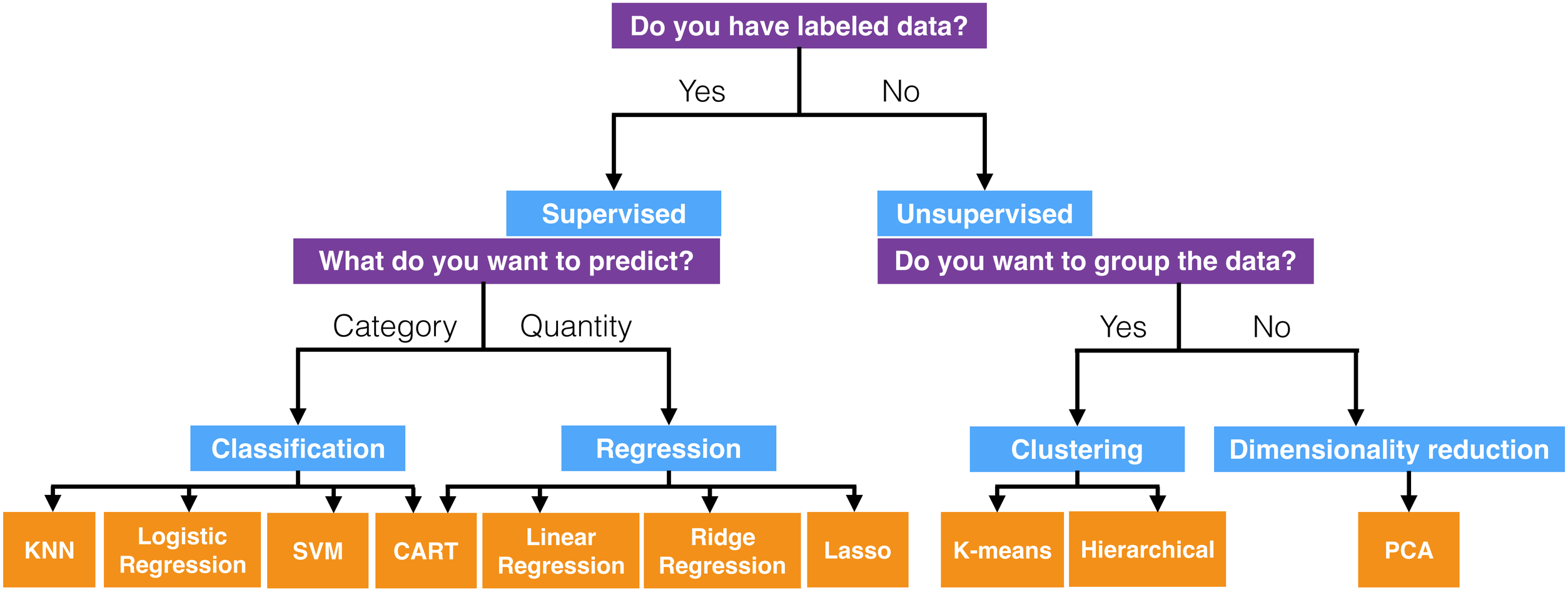
Agenda

Slides are online at
cme250.stanford.edu

- Clustering methods
  - K-means clustering
  - Hierarchical clustering
- Dimensionality reduction
  - PCA

Machine Learning Methods



Unsupervised Learning

Recall: A set of statistical tools for data that only has features/input available, but no response.

In other words, we have X's but no labels y.

Goal: Discover interesting patterns/properties of the data.

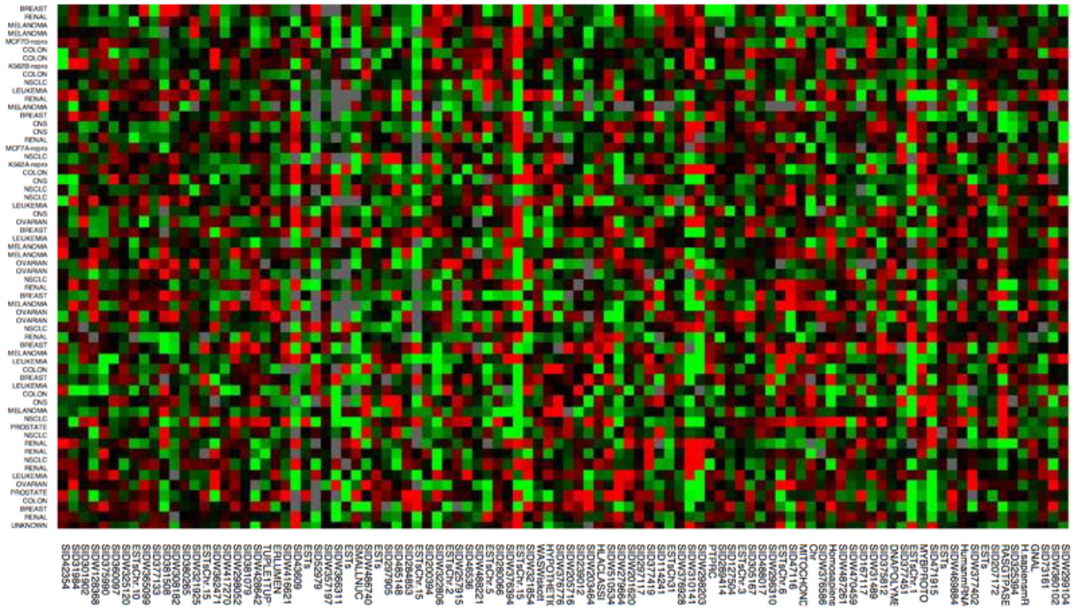
- E.g. for visualizing or interpreting high-dimensional data.

Unsupervised Learning

Example applications:

- Given tissue samples from n patients with breast cancer, identify unknown subtypes of breast cancer.

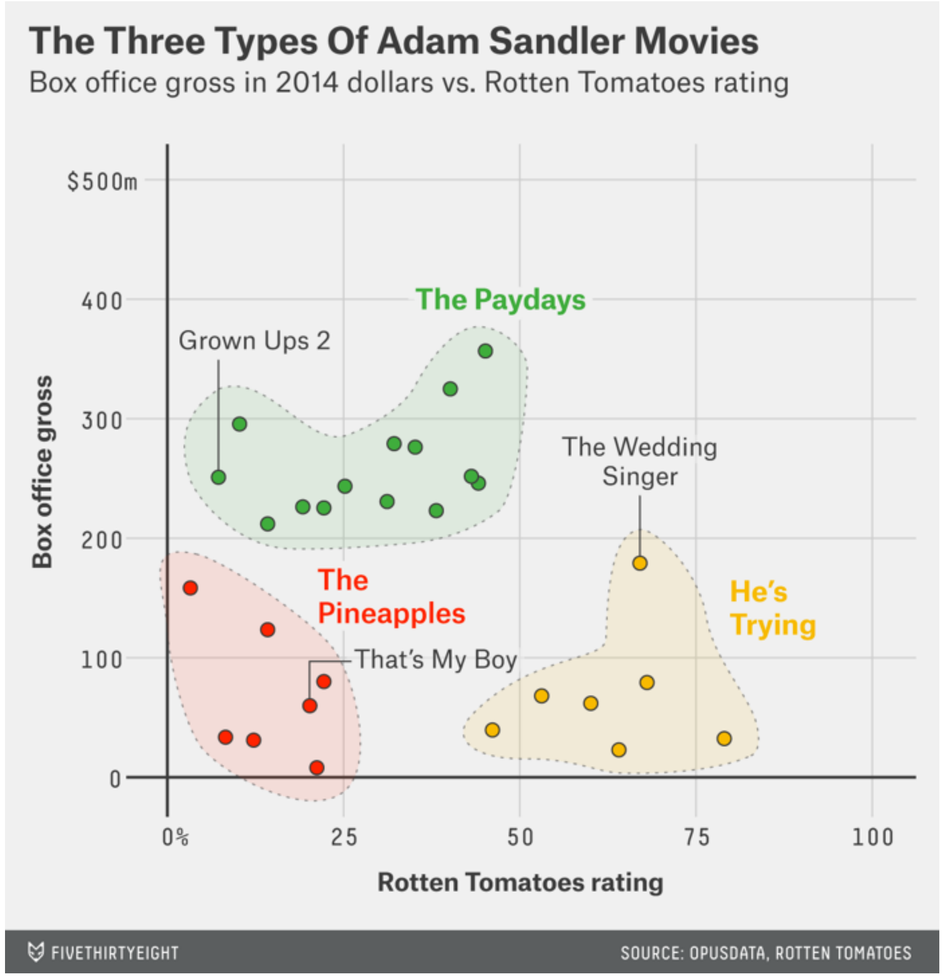
- Gene expression experiments have thousands of variables. Represent the data using a smaller set of features for visualization and interpretation.



Unsupervised Learning

Example applications:

- Document clustering: identify sets of documents about the same topic.
- Given high-dimensional facial images, find a compact representation as inputs for a facial recognition classifier.



Challenges of Unsupervised Learning

Why is unsupervised learning challenging?

- Exploratory data analysis — goal is not always clearly defined
- Difficult to assess performance — “right answer” unknown
- Working with high-dimensional data

Types of Unsupervised Learning

Two approaches:

- **Cluster analysis**
  - For identifying homogenous subgroups of samples
- **Dimensionality reduction**
  - For finding a low-dimensional representation to characterize and visualize the data

Cluster Analysis

Clustering

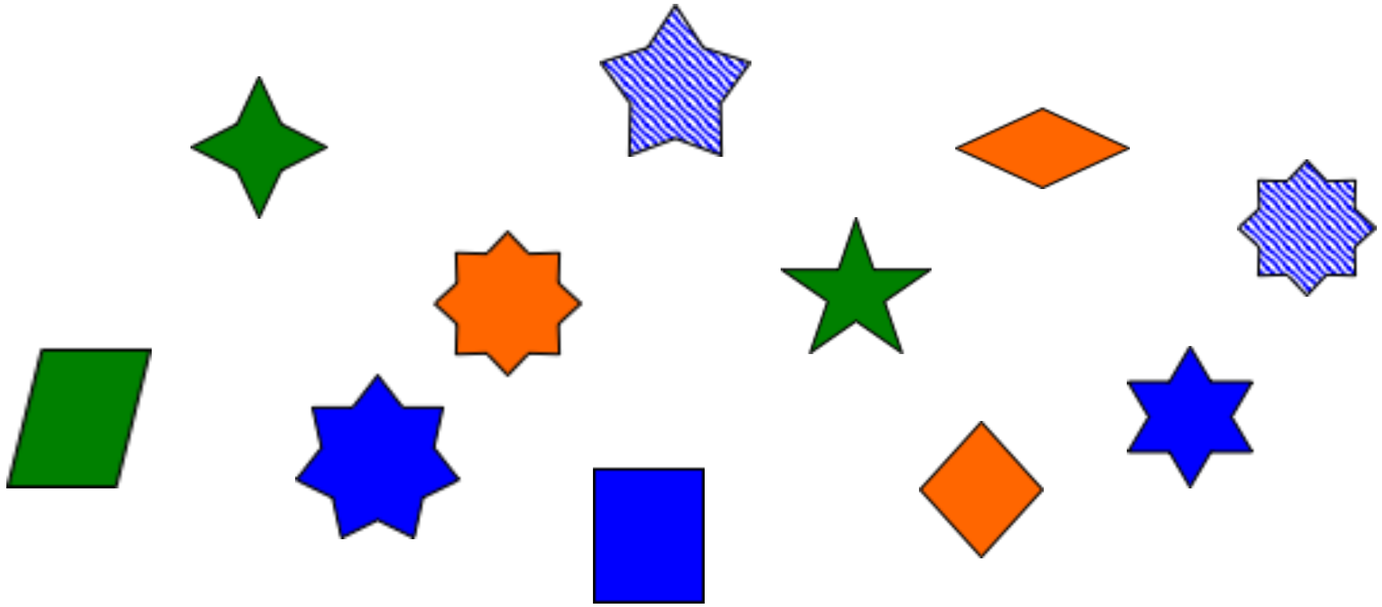
A set of methods for finding subgroups within the dataset.

- Observations should share common characteristics within the same group, but differ across groups.

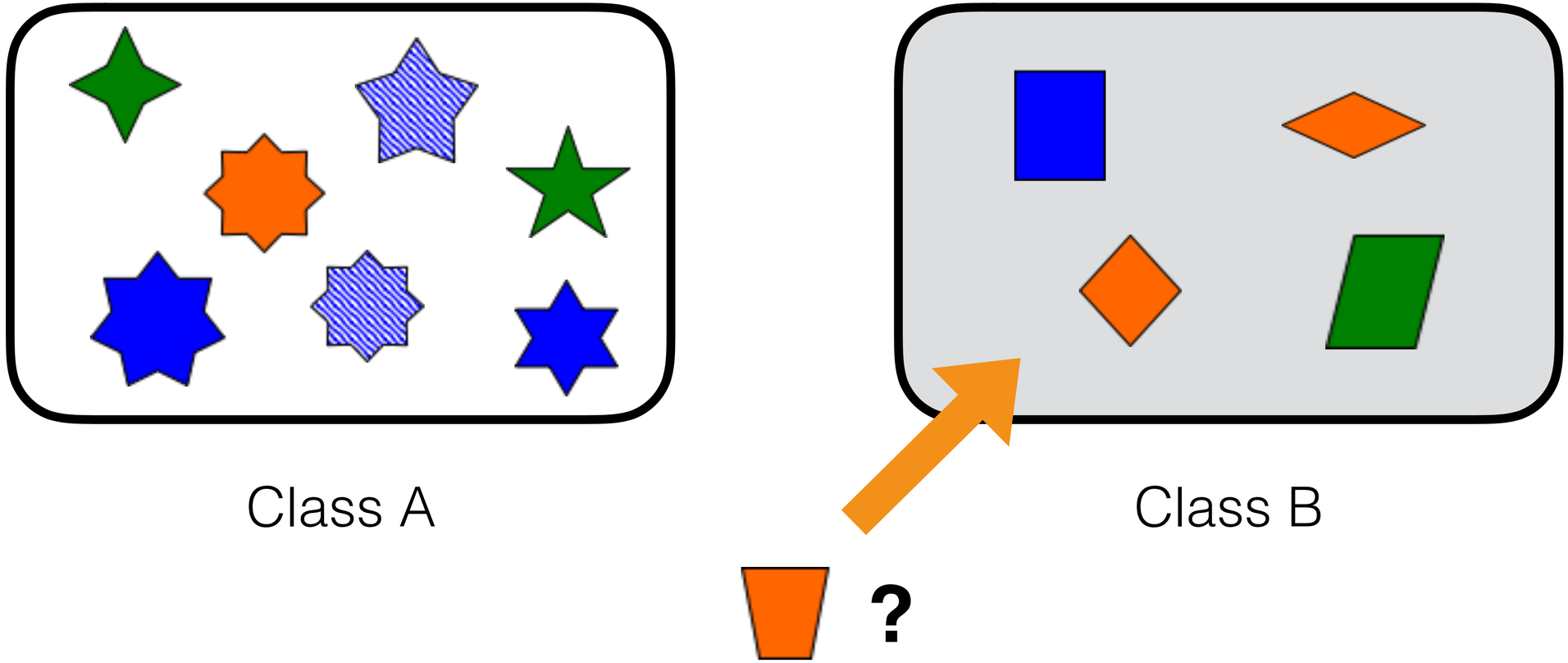
- Groupings are determined from attributes of the data itself — differs from classification.

https://medium.com/square-corner-blog/so-you-have-some-clusters-now-what-abfd297a575b

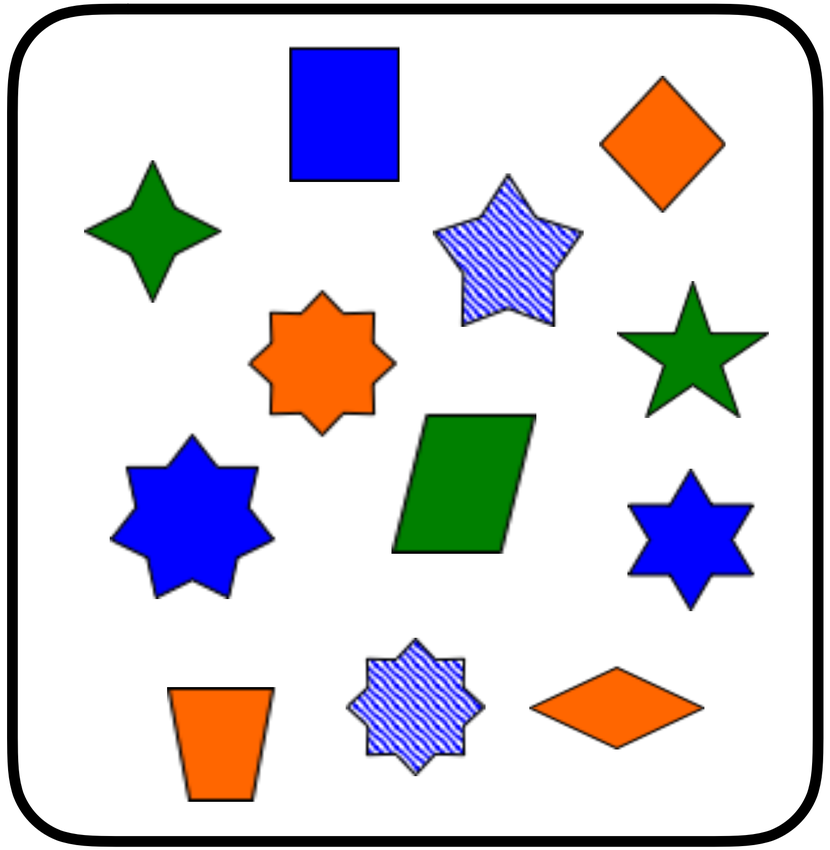
Clustering vs. Classification



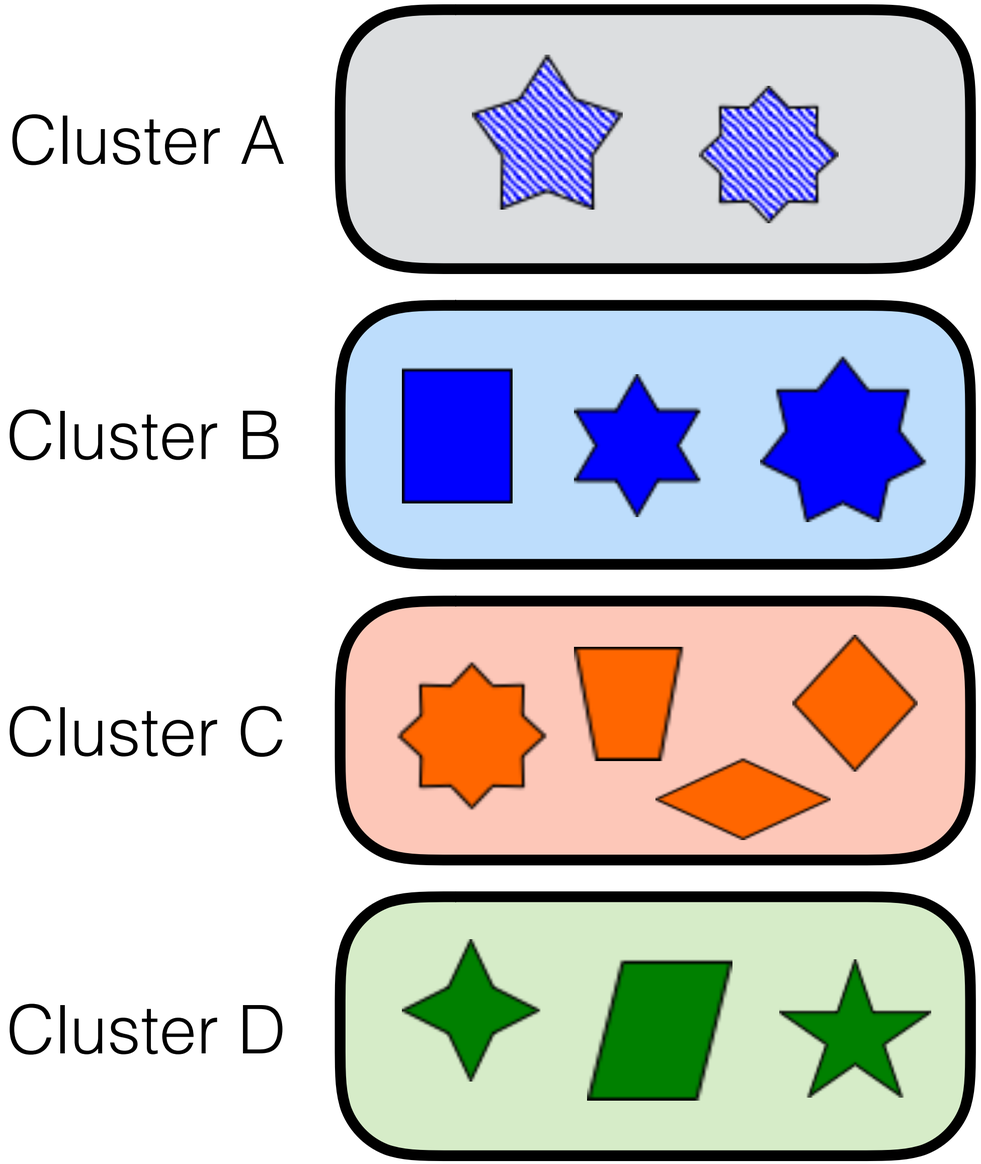
Classification



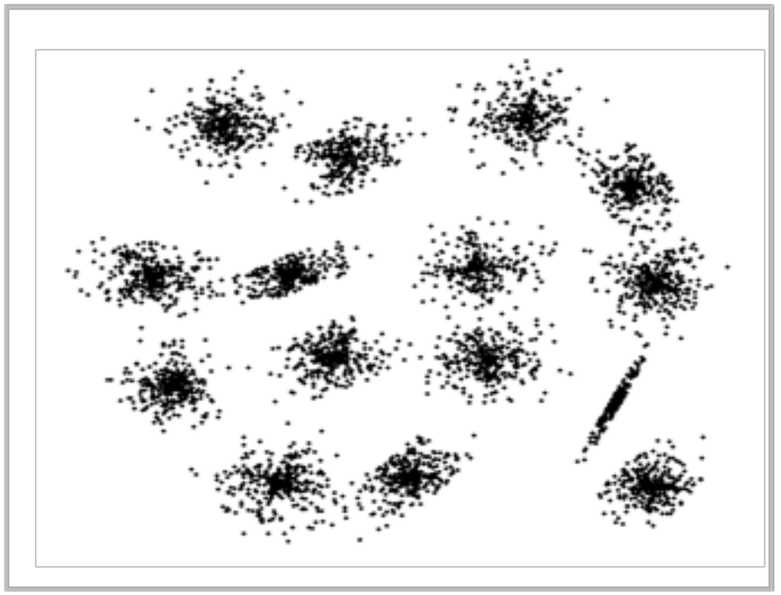
Clustering



Dataset



Clustering



http://cs.joensuu.fi/sipu/datasets/

Types of Clustering

- **Centroid-based clustering**
- **Hierarchical clustering**
- **Model-based clustering**
  - Each cluster is represented by a parametric distribution
  - Dataset is a mixture of distributions
- **Hard vs. soft/fuzzy clustering**
  - Hard: observations divided into distinct clusters
  - Soft: observations may belong to more than one cluster

Dimensionality Reduction

Dimensionality Reduction

Recall the curse of dimensionality when working in high dimensions.

Dimensionality reduction is the process of reducing the number of features under consideration.

We already saw some examples of this in the lasso and forward/backward selection algorithms. These methods reduce dimensionality by selecting a subset of features. However, they do so using supervision — knowing a response y that is of interest.